



Architecture Design Methodology on Huawei Cloud



Foreword

- Deploying service systems on the public cloud has become common across industries. To fully tap into the features of a cloud platform and ensure successful deployment, selecting a suitable architecture is a vital step. This lesson describes the basics of cloud computing, Huawei Cloud infrastructure, and architecture design methodology on Huawei Cloud, and explains a typical case to help you foster a basic understanding of cloud service deployment.



Objectives

- Upon completion of this course, you will be able to describe:
 - Basic concepts of cloud computing
 - Huawei Cloud infrastructure
 - Major driving forces for service cloud migration
 - Five principles of the cloud service architecture design

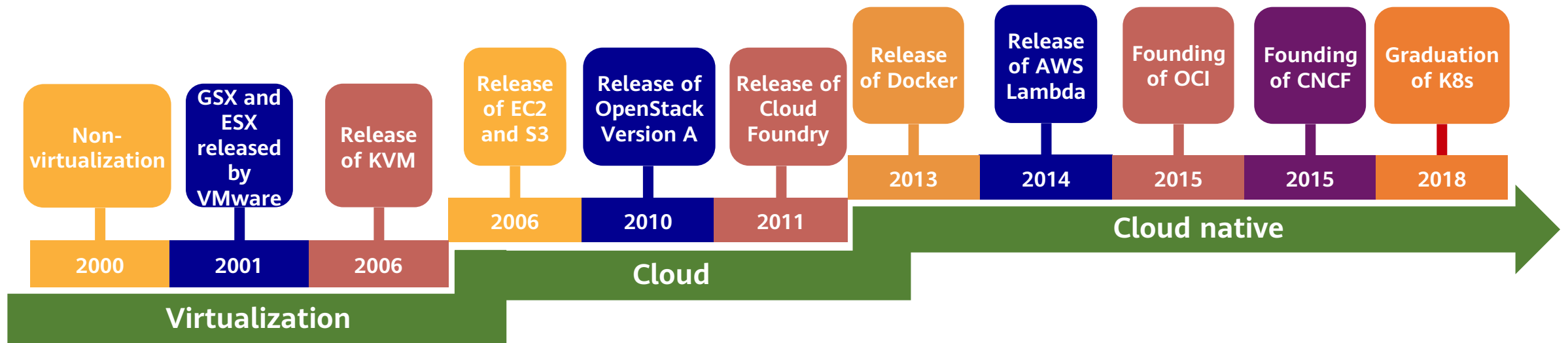


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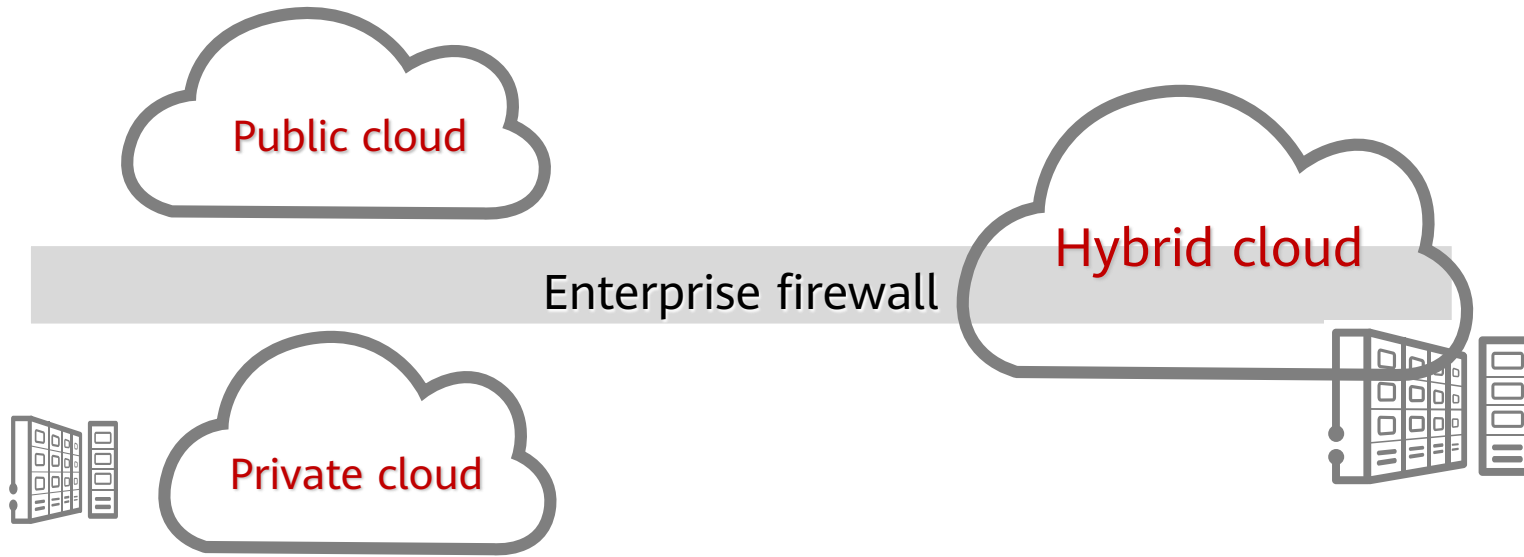
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Cloud Basics and Development Timeline

- Cloud computing refers to providing compute resources through on-demand services over the Internet. Enterprises do not need to purchase, configure, or manage resources on their own, and only need to pay for the resources they use.



Cloud Computing Development Models



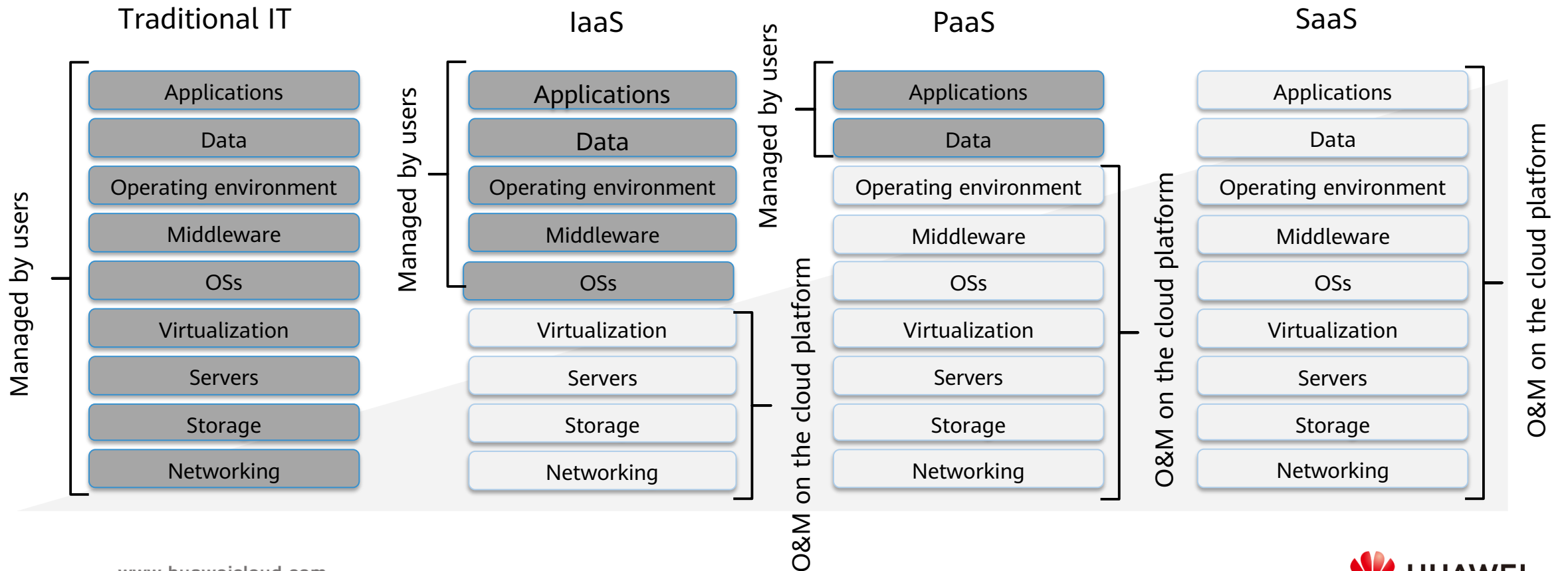
Private cloud: The cloud infrastructure is owned and managed for exclusive use by a single organization.

Public cloud: The cloud service provider owns and operates the cloud infrastructure and provides cloud services open to the public or enterprise customers.

Hybrid cloud: A combination of public and private clouds, viewed as a single cloud externally.

Cloud Computing Service Models

- There are three service models for cloud computing: Infrastructure as a Service (IaaS) that provides compute and storage services, Platform as a Service (PaaS) that provides environments for developing and deploying cloud applications, and Software as a Service (SaaS) that provides applications as services.

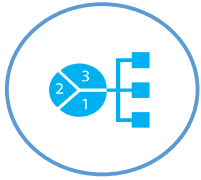


Cloud Computing Benefits



Reduced investment into fixed assets

- Investments into fixed assets are reduced in the early stage so that enterprise resources can be used to tap into the core value of businesses.
- Fixed assets may become a burden for future transformation. For example, business adjustments may be limited by the reuse of existing assets.



Large-scale resources

- Economies of scale, that is, the procurement volume from Huawei is enormous and the price is lower.
- Huawei Cloud is capable of providing abundant resources for a large number of businesses. Therefore, the resource requirement of one single customer is generally easy to be met on Huawei Cloud. It can be assumed that resources are always available.



Quick resource acquisition

- Resources can be obtained in just a few minutes.
- Therefore, users can easily implement dynamic service capabilities. Obtaining resources for service innovation is also convenient.



No need for capacity guessing

- Resources can be quickly obtained, so they are there when needed.
- Due to the large-scale resources, it can be assumed that resources are always available.



Easier O&M

- More automated and less manual O&M
- More application-level and less infrastructure-level O&M



Operations at different geographical locations

- The same standardized services are used to provide infrastructure.
- API-based access allows resources to be obtained in the same way.



Agile services

- Compute resources are quickly obtained to support agile services.
- A solid foundation is laid for companies to go cloud native and promote business innovation.

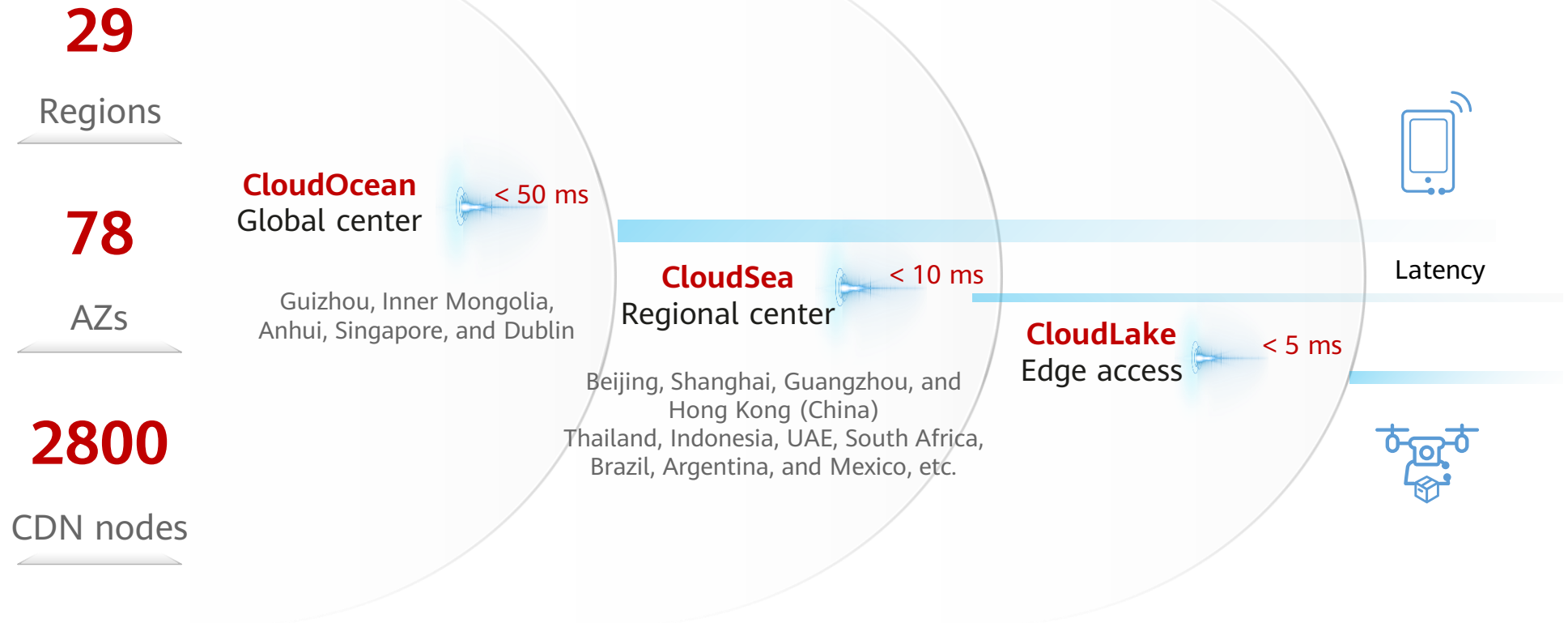
Focus on the core value of businesses and avoid repeated work



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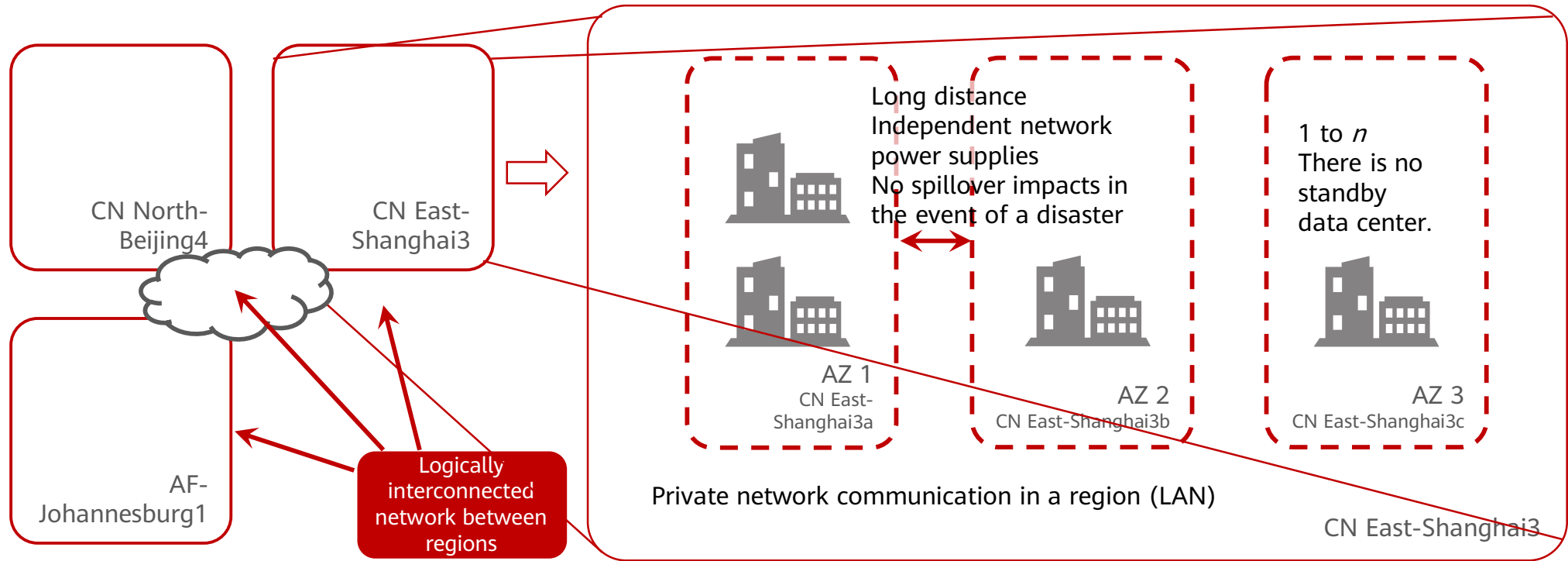
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Huawei Cloud's Global Presence for One-stop Services

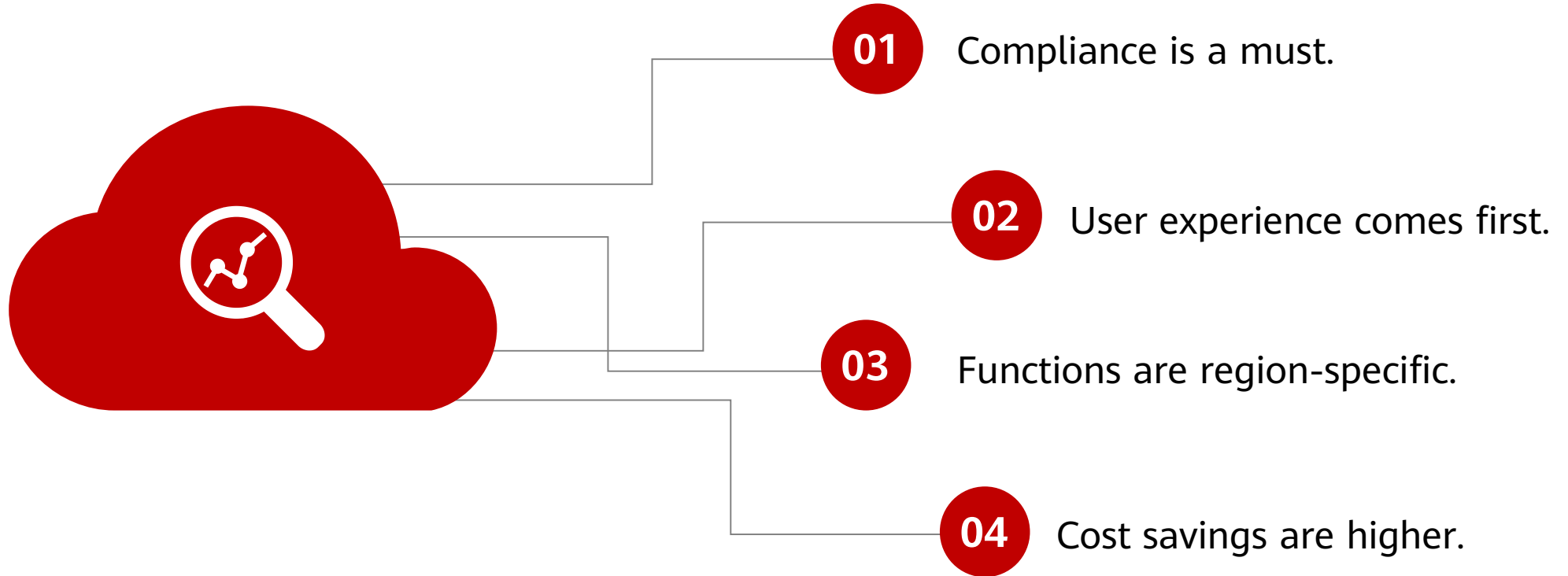


Huawei Cloud Infrastructure

- A region is a collection of resources divided by a geographic location. Each region comprises one or more AZs and is completely isolated from other regions. AZs within the same region can communicate with one another through a private network, while those in different regions cannot.
- An availability zone (AZ) is a geographical area with independent power supply and networks in the same service region. Generally, an AZ is an independent physical equipment room. One region has multiple AZs. If one AZ becomes faulty, the other AZs in the same region can still function. AZs within the same region can communicate with one another through a private network.



How Do I Select a Region?



How Do I Select an AZ?

AZ



1. AZs aim to improve availability.
2. AZs help build active-active and multi-active applications.
3. AZs are equivalent.

Application requirements prevail in the selection.

If there is no special requirement, one AZ is sufficient.

If high availability (HA) is required, select at least two AZs.

One more AZ does not increase costs.

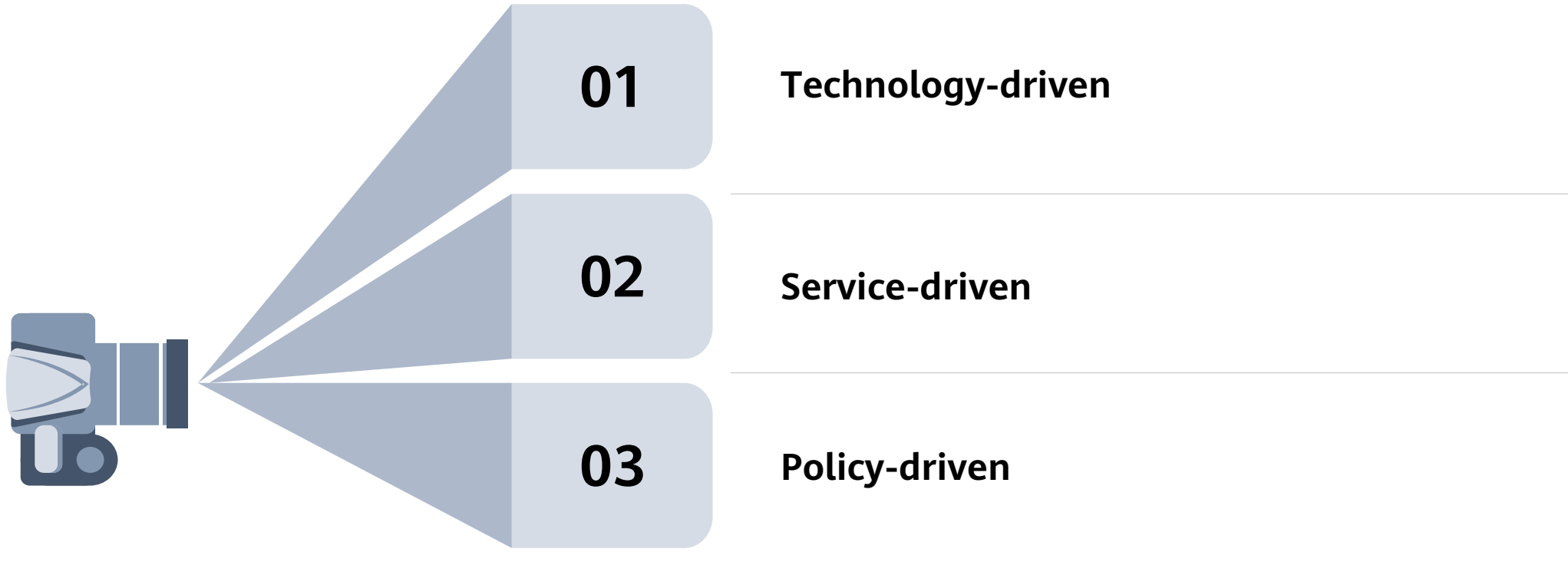
If there are more than three AZs, the availability improvement is limited and the complexity increases.



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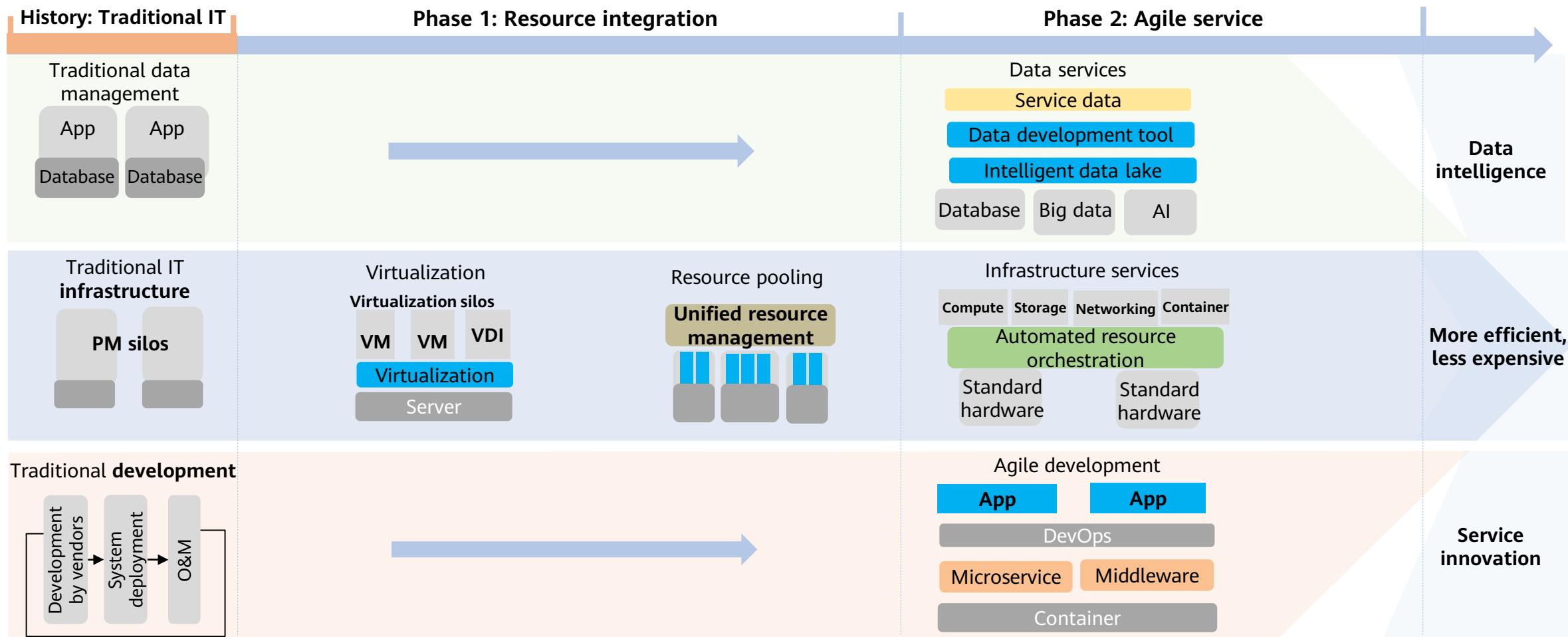
Major Driving Forces for Service Cloud Migration



An open, flexible, easy-to-use, and secure cloud platform provides enterprises with a new choice in modernizing their IT architectures.

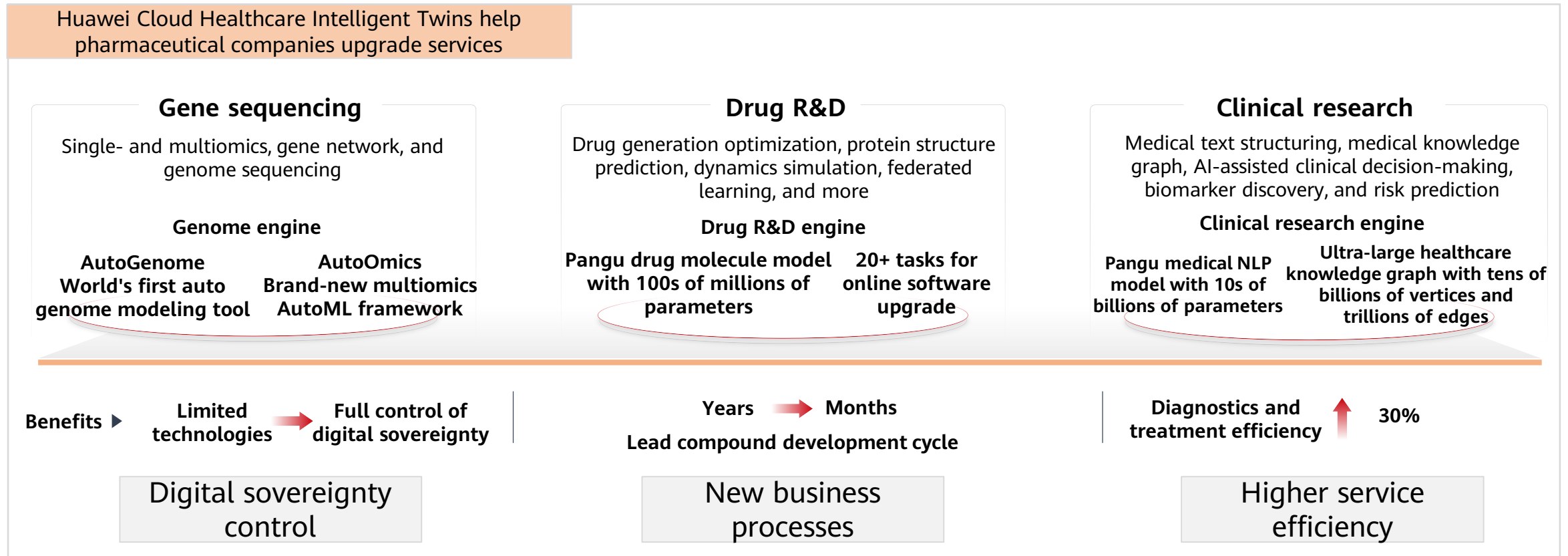
Technology-driven

- Cloud migration is essential for enterprises to achieve agile service development, process massive amounts of data, and build highly resilient IT systems.



Service-driven

- Cloud vendors provide a large number of PaaS and SaaS services and complete solutions to help enterprises explore new service requirements and enhance competitiveness.



Policy-driven

- Cloud computing is a key industry, and enterprises moving to the cloud are often supported and promoted by various governments worldwide.

*13th Five-Year Science and
Technology Innovation Plan*

2016 China

*Federal Cloud Computing
Strategy*

2019 USA

Shaping Europe's Digital Future

2020 Europe

*Outline of the 14th Five-Year Plan for
National Economic and Social
Development and Long-Range
Objectives Through the Year 2035*

2021 China

More



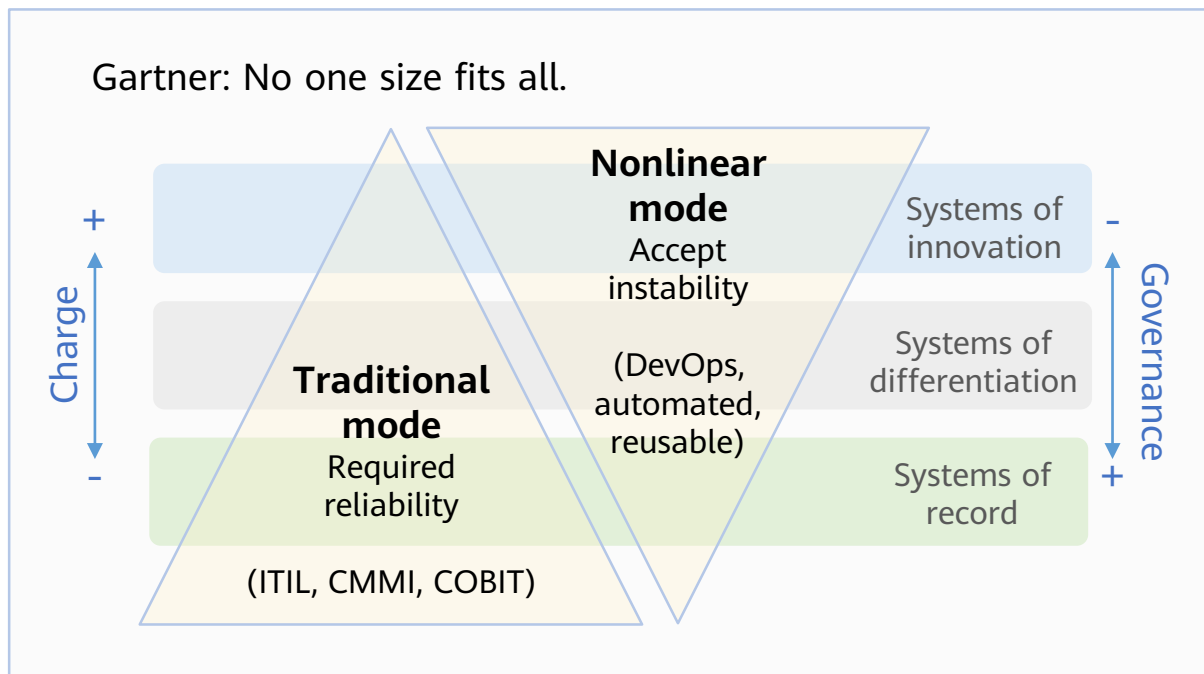


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Enterprise Cloud Architecture Design Requirements

The enterprise cloud architecture must meet the requirements of high reliability, high security, consistency, and compliance in enterprise IT, and be capable of flexible, fast, and scalable IT innovation.



1. Performance

Enterprises are eager for high-performance applications and fast analysis to quickly acquire information.

2. Cost

Enterprises are keen to optimize the scheduling of O&M personnel and reduce the manpower input in IT maintenance so that more resources can be allocated to promote the combination of service demands and IT technologies, motivating enterprise innovation.

3. Security

Enterprises require a complete security deployment with a strong focus on the security of data centers (physical infrastructure), access, networking, storage, and data.

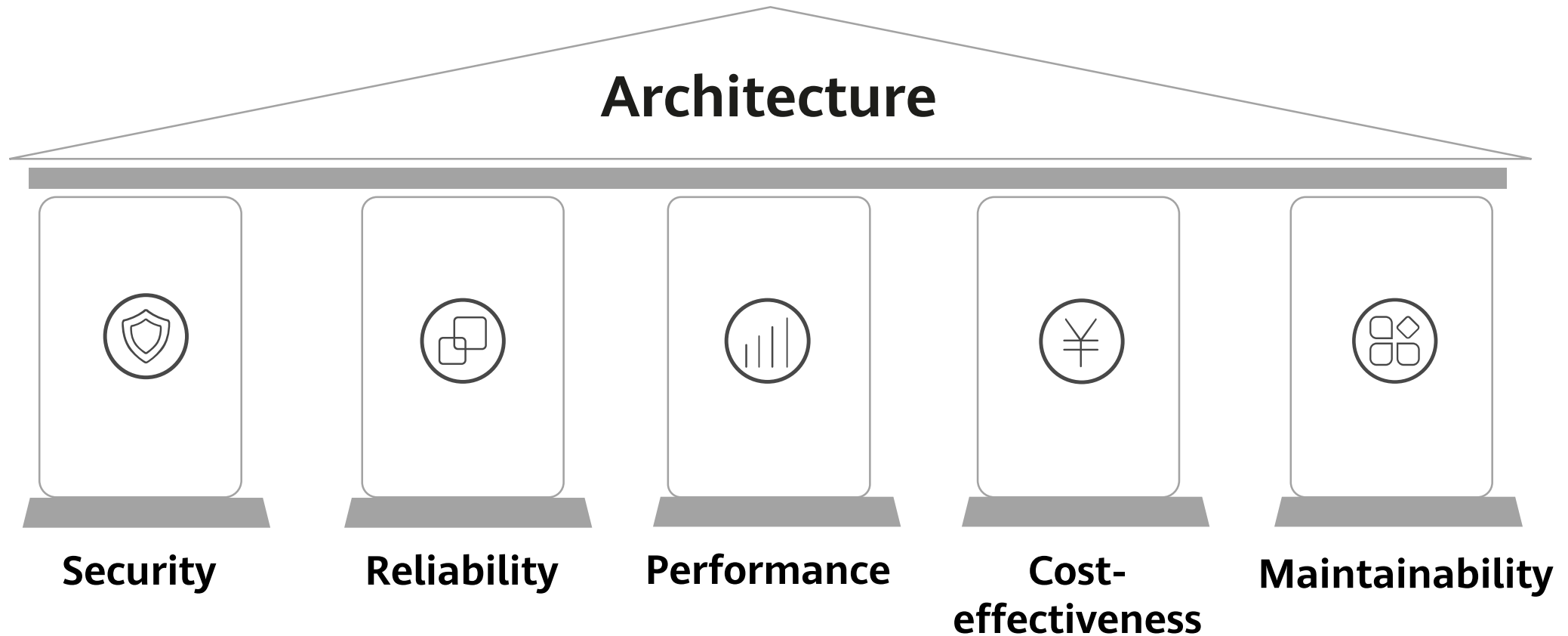
4. Availability

When a disaster occurs and one data center is faulty, the other data center can run in real time to ensure service continuity.

5. Maintainability

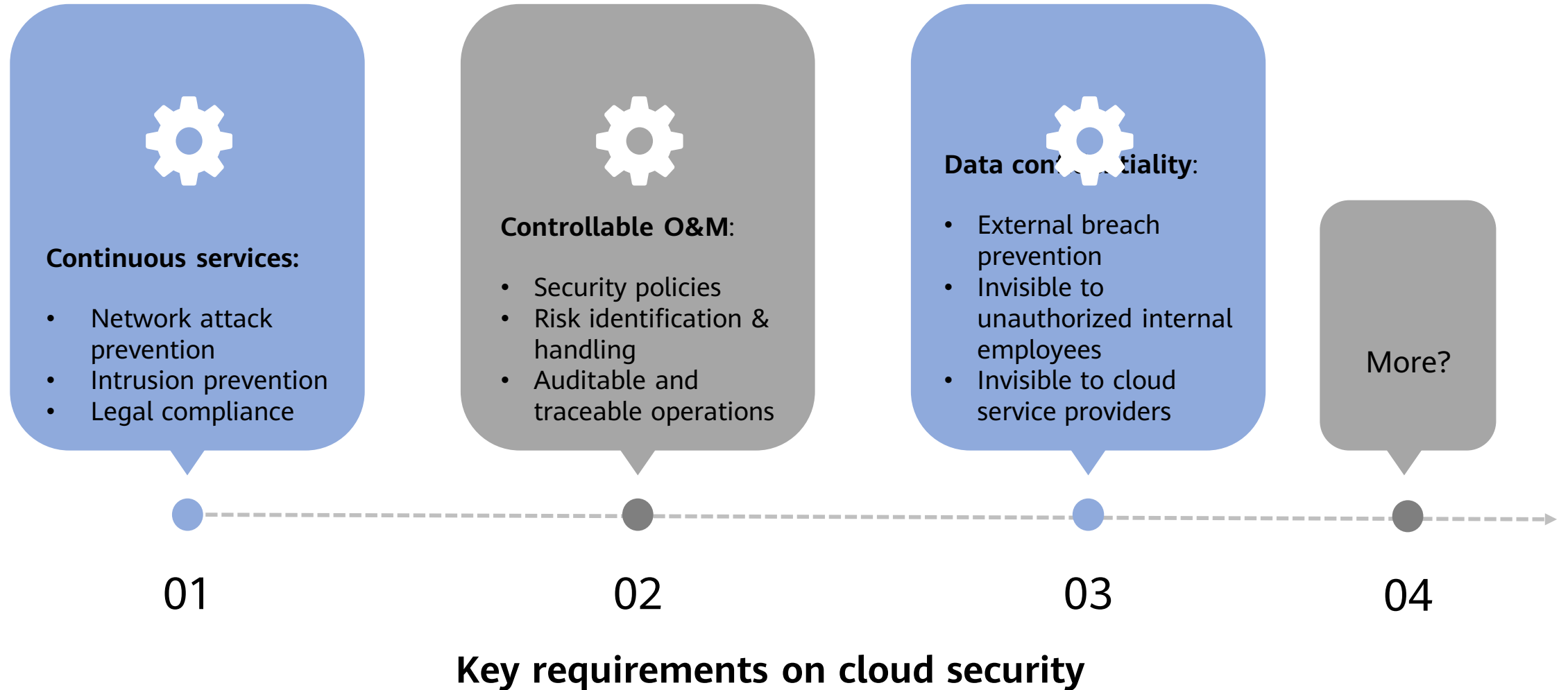
When a fault or risk occurs, services can be recovered or switched over as scheduled in a timely manner to ensure service continuity.

Five Principles of Solution Architecture Design



These five principles are used throughout the process of solution architecture design.

Cloud Security Customer Requirements

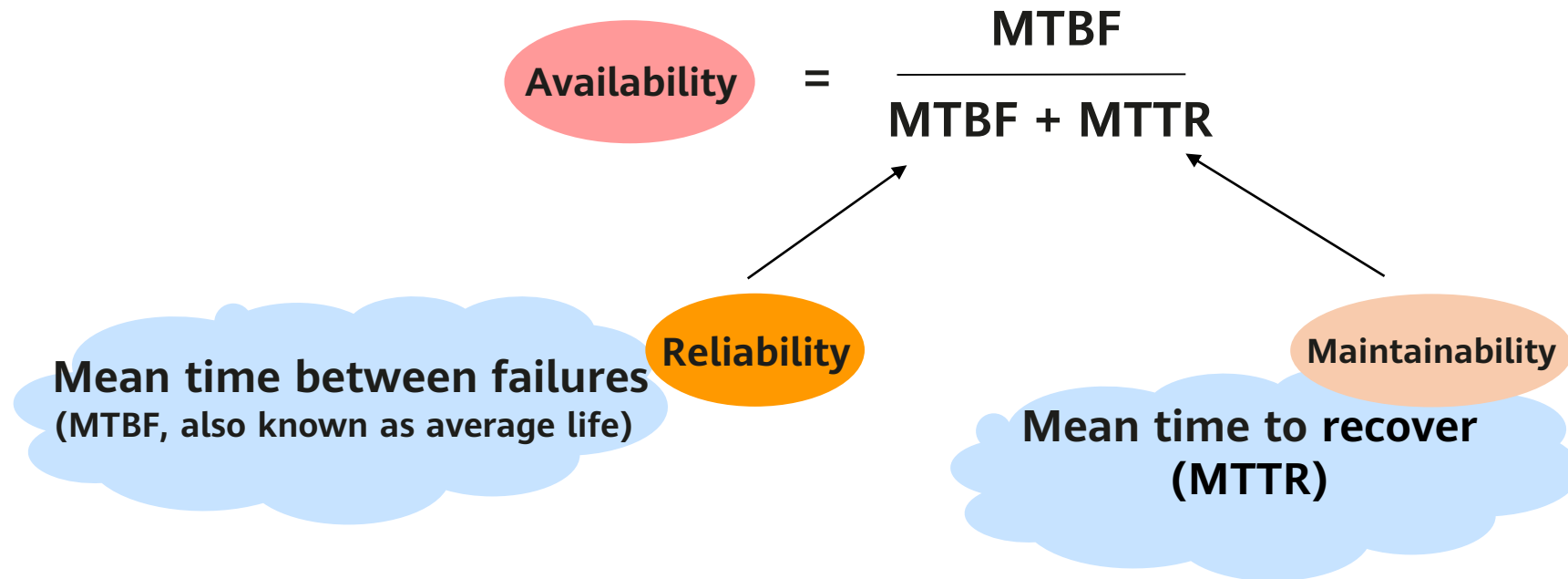


Cloud System Security Design Principles



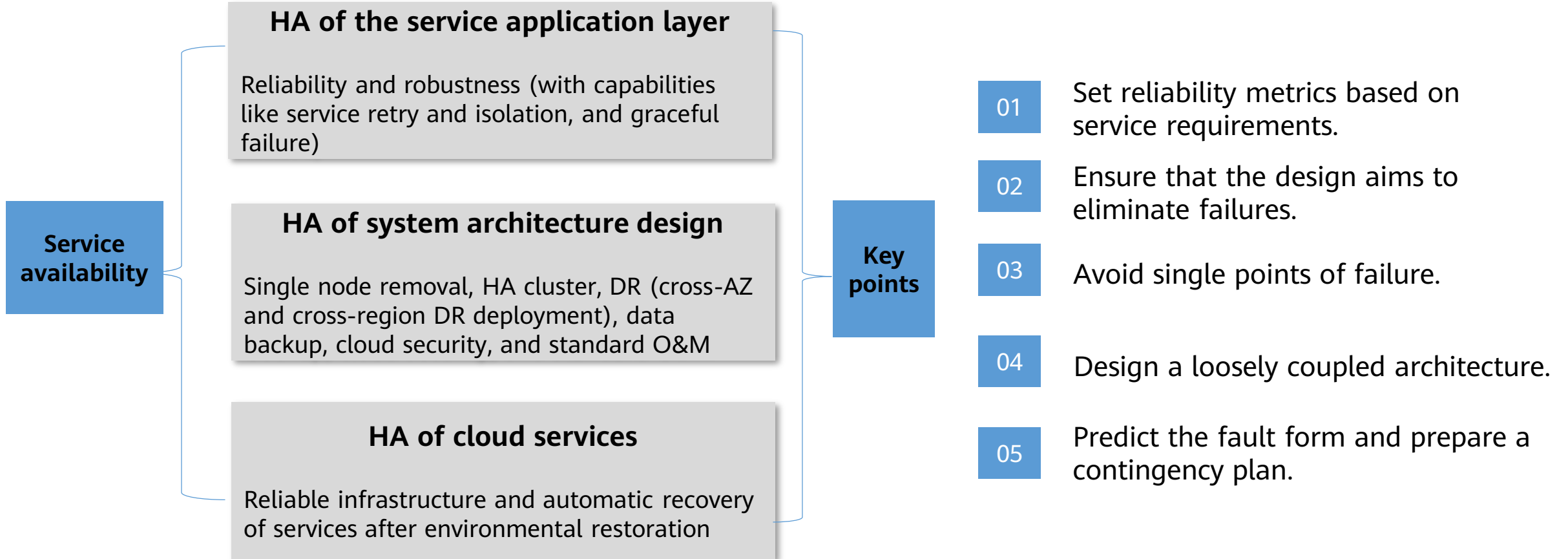
High-Availability Requirements

- The availability of a system is reflected in two aspects: **reliability** (service continuity) and **maintainability** (quick service restoration).



- Availability can also be regarded as service continuity. Instance availability = $(1 - \text{Fault duration} / \text{Total service duration}) \times 100\%$. For example, 99.995% availability means that a service can be accessed for 100,000 consecutive minutes (about 69 days), during which the service is unavailable for no more than five minutes.
- Reliability refers to the probability that products are able to execute the specified functions as required in the specified conditions over the specified time.

Cloud System High-Reliability Design Principles



Cloud System Performance Metrics

Compute

Latency refers to the time it takes to start performing an operation. Latency is the most intuitive metric of cloud computing performance.

Networking

Throughput is a metric that evaluates data processing speed. In terms of data transmission, throughput is expressed in bytes/second or bits/second. Throughput limit is one of the common performance bottlenecks.

Storage

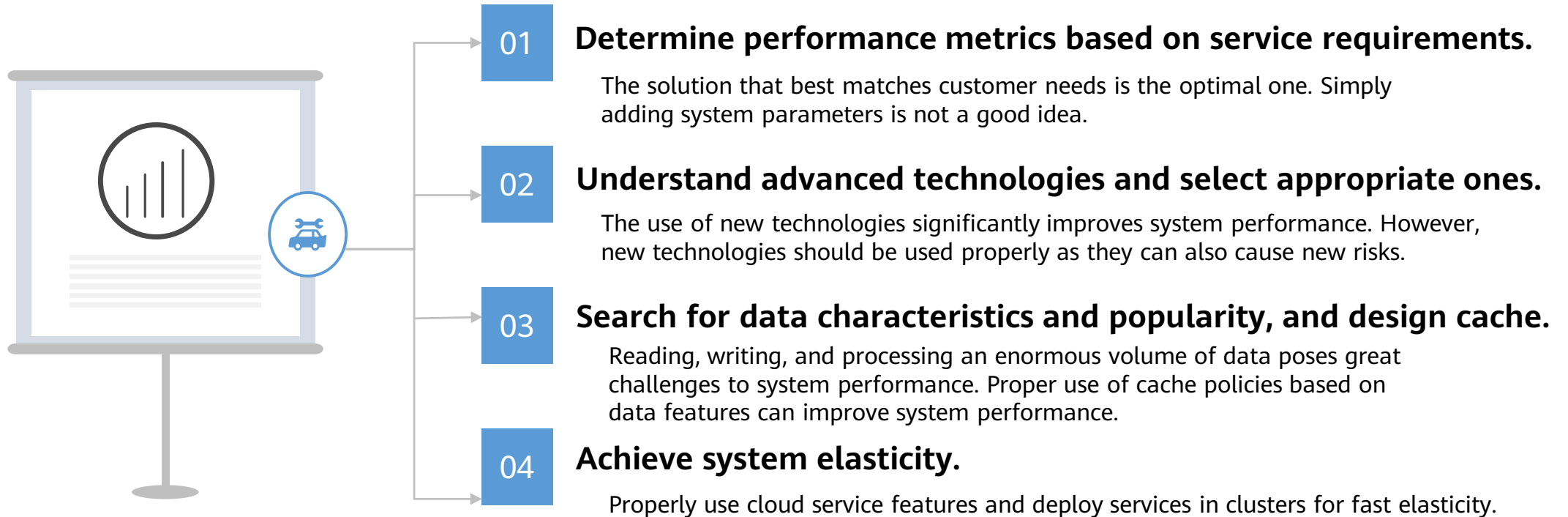
IOPS refers to the number of input/output operations per second. It is a measurement of data transmission.

Database

Concurrency refers to the capability of running several programs simultaneously within a period of time.

Latency, throughput, IOPS, and concurrency are key factors related to cloud application performance.

Cloud System Performance Design Principles

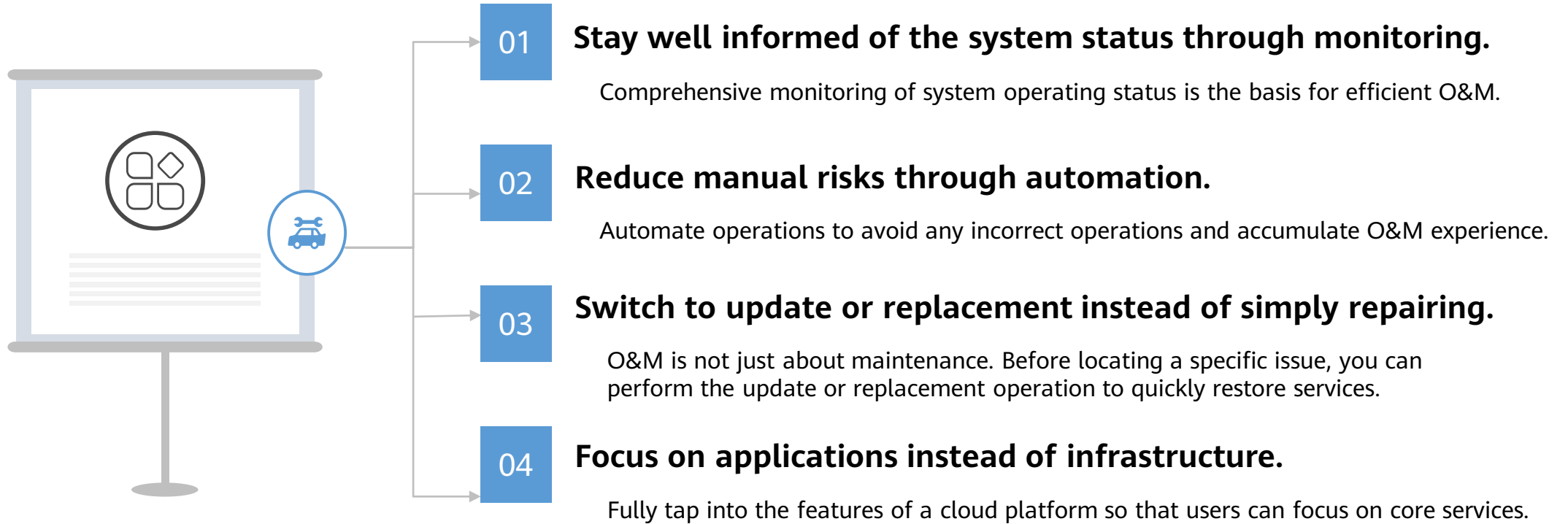


The performance does not need to be extreme. Being slightly higher than the standard requirements is sufficient.

Cloud System Cost Optimization Design Principles

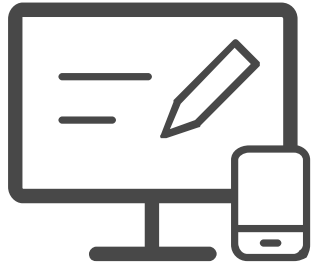
- 01 **Continuously reducing costs:** Allocate costs based on the system resource utilization over time.
- 02 **Using cloud hosting services:** Customize a plan to pay only for what you use.
- 03 **Balancing supply and demand:** Avoid over-provisioning to eliminate unnecessary costs.
- 04 **Fostering an awareness of expenditure:** Accurately track costs and assess the profitability of different departments and products.
- 05 **Using cost-effective resources:** Cost-effective resources and appropriate services and specifications are essential to lowering costs.
- 06 **Optimizing costs at multiple levels:** Optimize costs from multiple aspects.

Cloud System Maintainability Design Principles



A good architecture should be easy to implement and maintain.

When Can Architecture Design Principles Come into Play?



Architecture design



Architecture acceptance

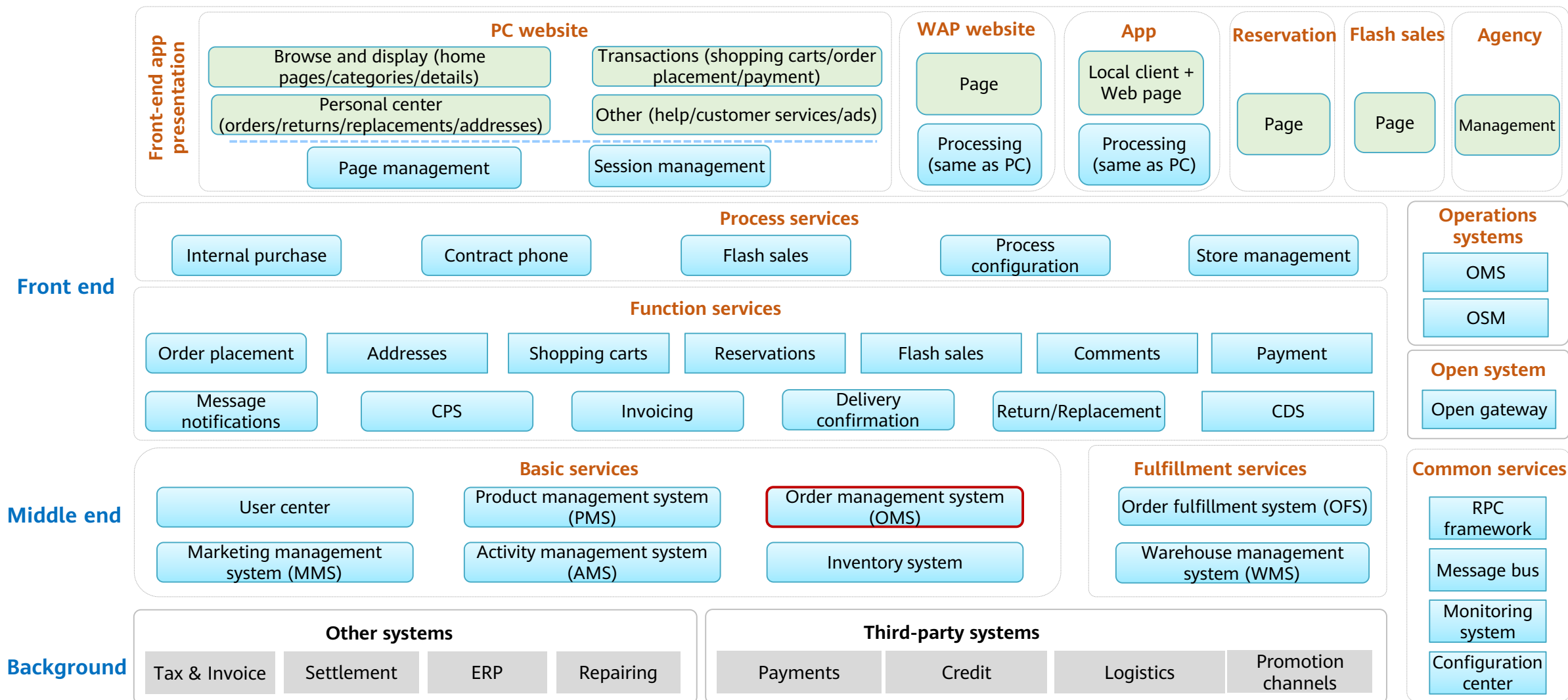
Follow architecture design principles to solve pending issues.



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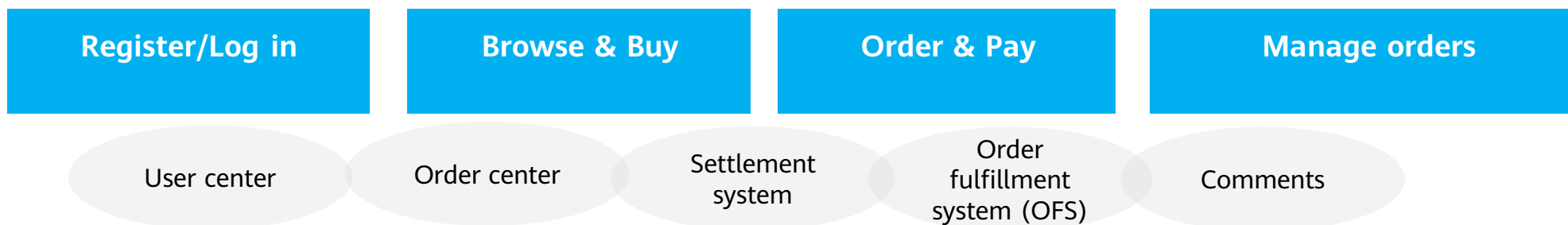
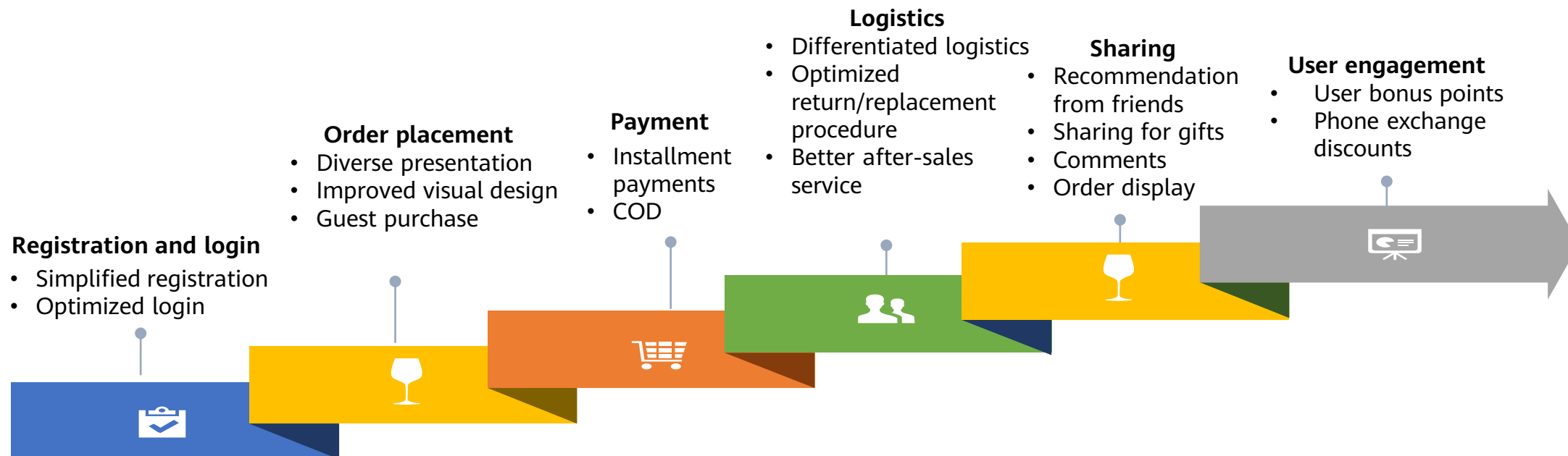
Vmall Target Service Architecture



Analysis of Vmall Target Service Architecture

- Vmall is a Huawei-operated e-commerce platform that looks to facilitate lives with mobile information. Typical activities of Vmall include device sales, holiday promotions, and new device launches, which are supported by Huawei Cloud services.
 - The front-end application presentation defines the touch points that allow users to interact with the system.
 - Front end
 - Process services: A complete product sales workflow of Vmall includes five core service processes.
 - Function services facilitate purchase for better experience.
 - Middle end
 - Basic services are core functions.
 - Fulfillment services include order fulfillment and warehousing management.
 - Backend
 - The terms "other systems" and "third-party systems" indicate the areas where Vmall integrates with other systems.
 - Open system: External participants will perform operations by calling APIs.
 - Message communication: The RPC framework and message bus are used for a distributed architecture. The RPC framework is used for synchronous communication between applications to ensure effective management. Asynchronous messages are transmitted through the message bus to ensure performance and reliability.

Vmall Service Process



Vmall Service Scenario: Holiday Promotion & New Product Release

Massive access requests and high concurrency

1. Performance: concurrent online purchasing for millions of users
2. Order placement: hundreds of thousands of orders per second
3. Payment: tens of thousands of online transactions per second
4. Service volume: 5 to 10 times (100 times in certain scenarios) normal volume

Security & reliability

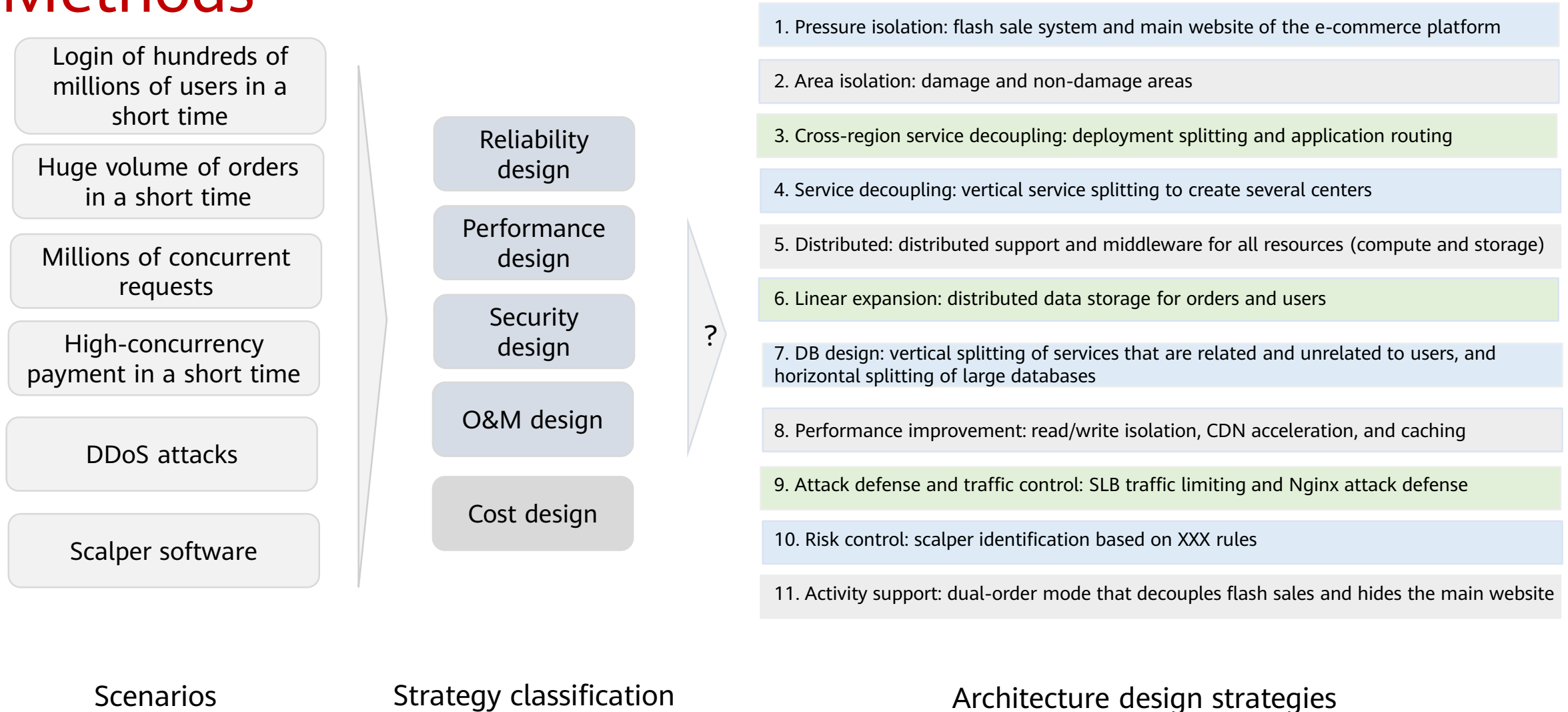
1. Robust defense: Massive DDoS attacks can be prevented.
2. Scalper bot prevention: The system identifies scalpers and prevents them from using automated software-programmed bots to automatically purchase goods to ensure normal pricing and availability for legitimate users.

User experience

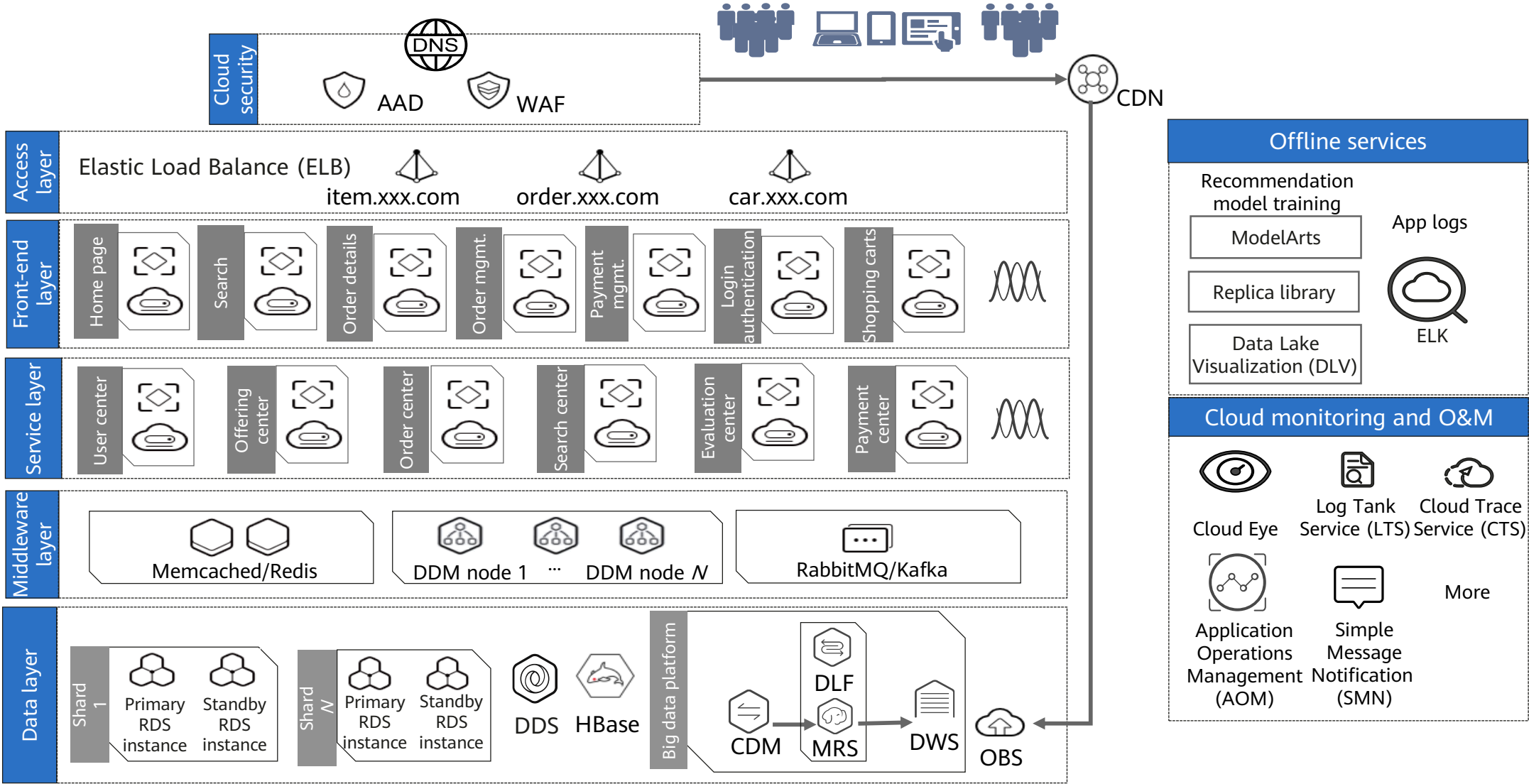
1. Reliable: smooth 24/7 access
2. Smooth: smooth operations without frame freezing at key time points

Holiday promotion (Double 11 shopping festival)/New product launch (Porsche) poses huge challenges to the performance, security, and reliability of cloud infrastructure.

Flash Sale Service Requirement Breakdown and Methods



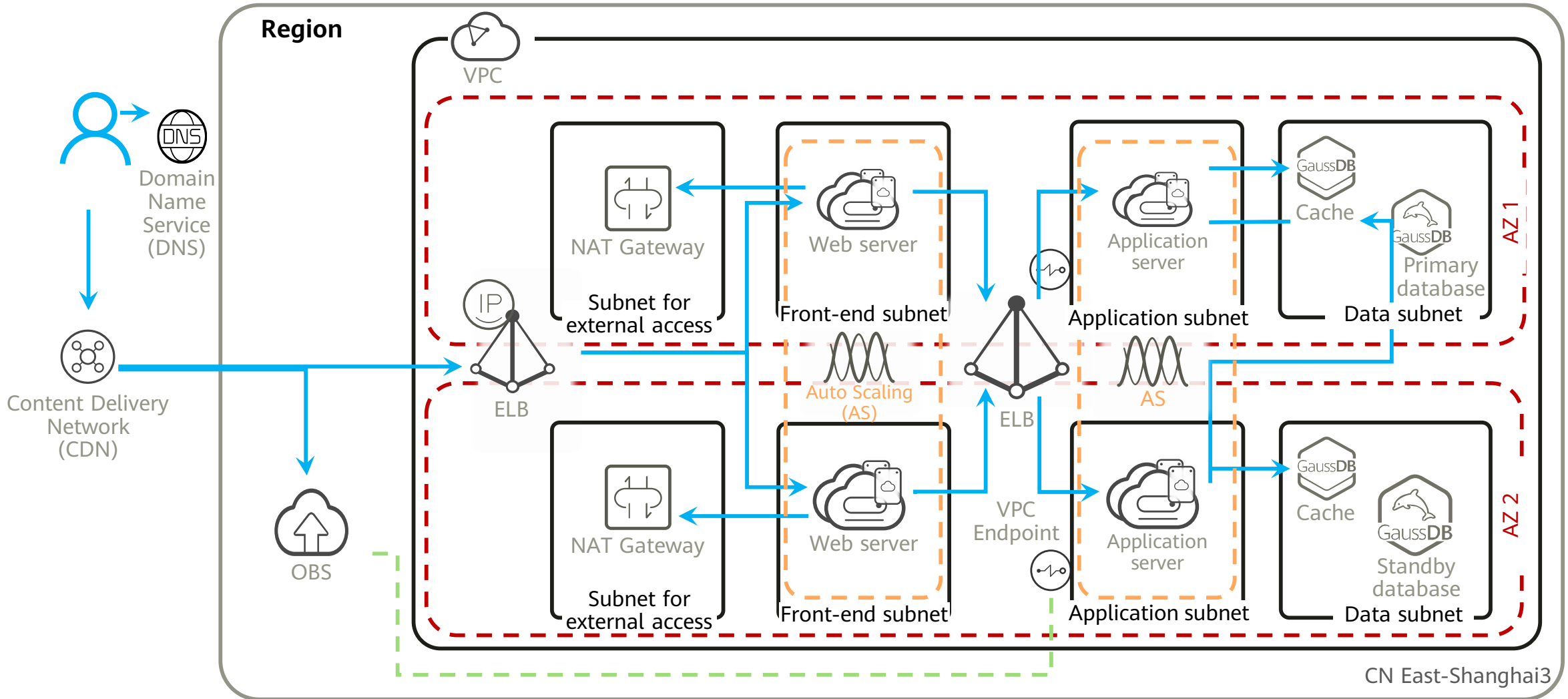
Cloud E-commerce Application Architecture



Interactive Tasks

How do we design a good cloud architecture to deploy the services of the order center?

Now you can understand this architecture:



Quiz

1. (Multiple-choice question) After a service system is deployed on Huawei Cloud, in which of the following aspects can we evaluate and optimize the service system on the cloud?
 - A. Reliability
 - B. Performance
 - C. Security
 - D. Maintainability



Acronyms and Abbreviations

- ECS: Elastic Cloud Server
- EVS: Elastic Volume Service

Thank You.

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